

2(a)(i)

any **four** from:

similarities

1 photoactivation of chlorophyll / AW, occurs in **both** ; **A** excite electrons (for AW)

2 ETC involved in **both** ;

3 ATP produced in **both** ;

differences

	cyclic	non-cyclic	
4	only PSI	PSI and PSII both involved	;
5	no, reduced NADP / oxygen, produced	reduced NADP / oxygen, produced	;
6	no photolysis or no oxygen-evolving complex involved	photolysis or oxygen-evolving complex involved	;
7	electrons emitted from PSI returned to PSI or PS1 is source of electrons	electrons emitted from PSII are replaced by water or water is source of electrons	;

Question	Answer	Marks														
2(a)(ii)	any two from: 1 no ATP and reduced NADP made ; 2 no, GP / TP, made or no, Calvin cycle / light-independent reaction ; 3 no regeneration of RuBP ;	2														
2(b)(i)	any four from: 1 as the CO₂ concentration increases, the rate of fixation of CO₂ increases ; 2 (as) CO₂ concentration is the limiting factor ; 3 as the CO₂ concentration increases, the rate of fixation of CO₂, remains the same / plateaus ; 4 (as) CO₂ concentration is no longer the limiting factor or temperature / light intensity / RuBP regeneration, is the limiting factor ; 5 paired data quote with units to support, mp1 / mp3 ; <table border="1"> <thead> <tr> <th></th><th>CO₂ concentration / mg m⁻³</th><th>CO₂ fixation rate / μmol CO₂ m⁻² s⁻¹</th></tr> </thead> <tbody> <tr> <td rowspan="2">mp1</td><td>50 ±10</td><td>1 ±0.25</td></tr> <tr> <td>1200–1280</td><td>42 A 41.75</td></tr> <tr> <td colspan="3"></td></tr> <tr> <td>mp3</td><td>from 1200–1910</td><td>42 A 41.75</td></tr> </tbody> </table>		CO ₂ concentration / mg m ⁻³	CO ₂ fixation rate / μmol CO ₂ m ⁻² s ⁻¹	mp1	50 ±10	1 ±0.25	1200–1280	42 A 41.75				mp3	from 1200–1910	42 A 41.75	4
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Question	Answer	Marks
2(b)(ii)	any three from: <i>assume Sox4 – accept ora for wild</i> 1 the rate of fixation of CO₂ is higher in Sox4 compared to wild type or Sox4 reaches a higher (maximum) rate of CO₂ fixation ; 2 (wild type) 42 vs (Sox4) 47 μmol CO₂ m⁻²s⁻¹ ; 3 (Sox4 has) more SBPase or there is a faster (rate of) regeneration of RuBP or (new) SBPase more effective ; 4 more RuBP to react with, CO₂ / rubisco ;	3